

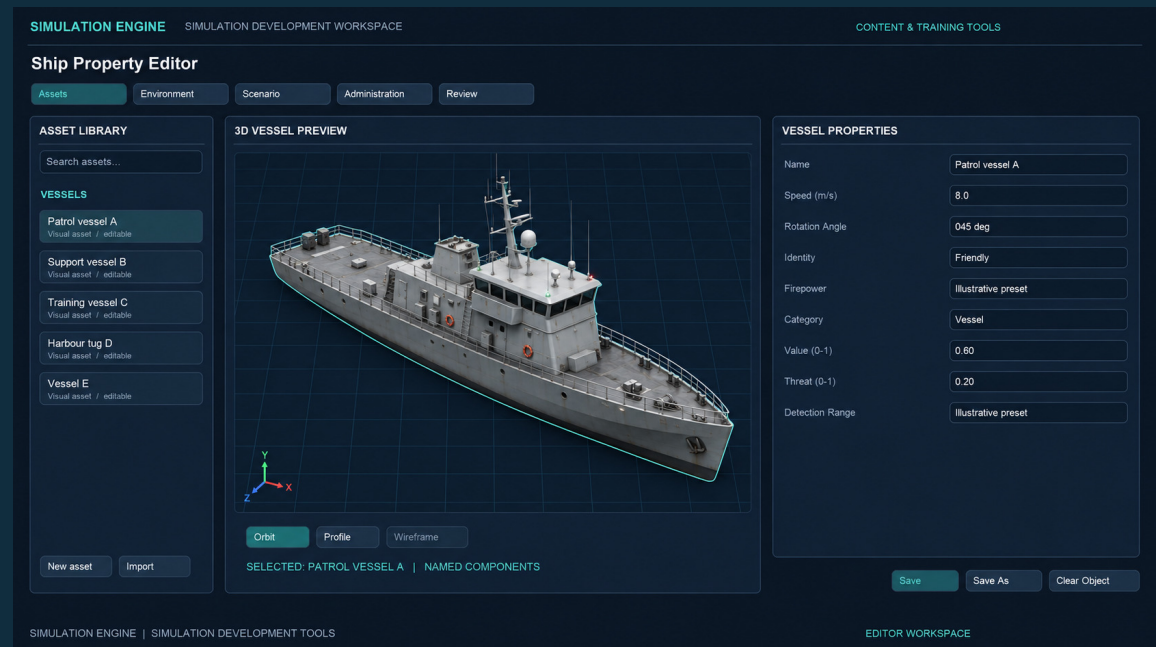
Simulation Development Portfolio

In-house development capability
and local delivery for RMN

Immersive training

Editable 3D and geographic content

Simulation Engine development tools



Simulation development portfolio



In-house development capability

VR Universal develops simulation software, editable 3D content, immersive training applications and physical control interfaces. Our company portfolio covers emergency response, command exercises, driving, drone training and naval simulation development.

Application to RMN

Our RMN approach brings together ship development, geographic area authoring, simulation content databases and development tools. Editable assets, structured records and configurable scenarios connect content creation with ongoing maintenance and local technical support.

Simulation development spanning immersive training, instructor workspaces and physical interaction.

Fire safety and command training



Coordinated training experience

Our Zhushan Training Center experience in Taiwan combines tabletop, VR and AR training with spatial workflows and software/hardware integration. The programme supports coordinated exercises for up to 20 participants.

Scenario and command development

Incident configuration, unit deployment and command workflows inform our instructor-led training approach. Editable scenarios and coordinated trainee roles provide a foundation for adapting simulation applications to maritime requirements.

Fire response and command training: incident configuration, coordinated roles and instructor-led exercises.

VR and physical interaction



Digital and physical interaction

Our portfolio includes vehicle driving with motion-based hardware, drone training with dedicated controllers and weather-recognition exercises. Training props and control interfaces connect the virtual environment with practical interaction.

Integration discipline

Our integration approach connects control mapping, user feedback and coordinated simulation states. It brings physical controls, tracking inputs and application behaviour together around the training task and operator workflow.

VR driving and physical interaction: motion hardware, control mapping and connected training displays.

Ship model development



Editable vessel development

Our vessel development study includes named components, editable Blender geometry and materials, and GLB export. The workflow connects visual modelling with asset references and vessel properties, organising content for simulator integration and subsequent modification.

Detail matched to requirements

Low, medium and high detail treatments show how vessel appearance can be scaled for the intended display and training task. Geometry, surface materials, viewing distance and runtime budgets guide asset preparation, allowing visual content to be tailored to project requirements.

Vessel content authoring with low, medium and high visual detail. Editable project: DEV_SHIP_001.

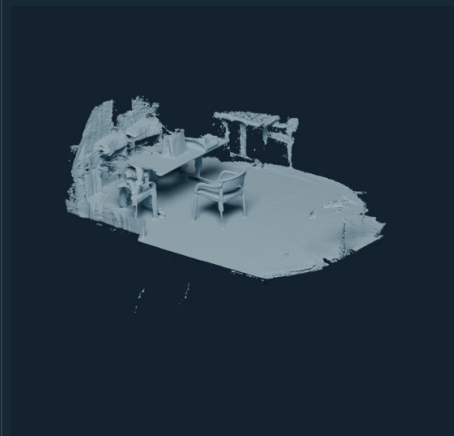
Captured content and optimisation

SIMULATION ENGINE / CAPTURED CONTENT DEVELOPMENT

Scan inspection and optimisation

From reconstructed geometry to an editable simulation asset

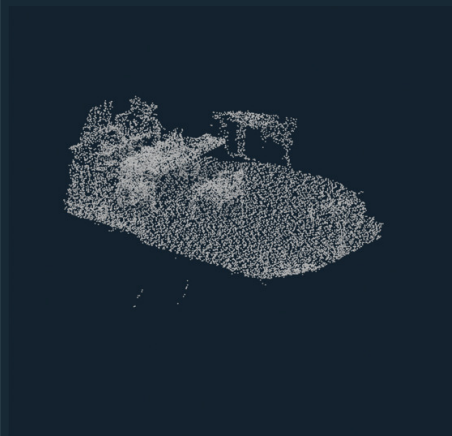
01 SOURCE RECONSTRUCTION



504,280 vertices / 935,255 faces

Public reconstructed scan mesh

02 POINT INSPECTION



Sampled source vertices

Derived inspection view

03 OPTIMISED MESH



139,968 faces / 85% reduction

Editable derivative and GLB export

Consistent camera angle across all stages. Point inspection is derived from mesh vertices.

Public source: Redwood Object Scans — Choi, Zhou, Miller and Koltun (2016), model 00021.

In-house processing demonstration

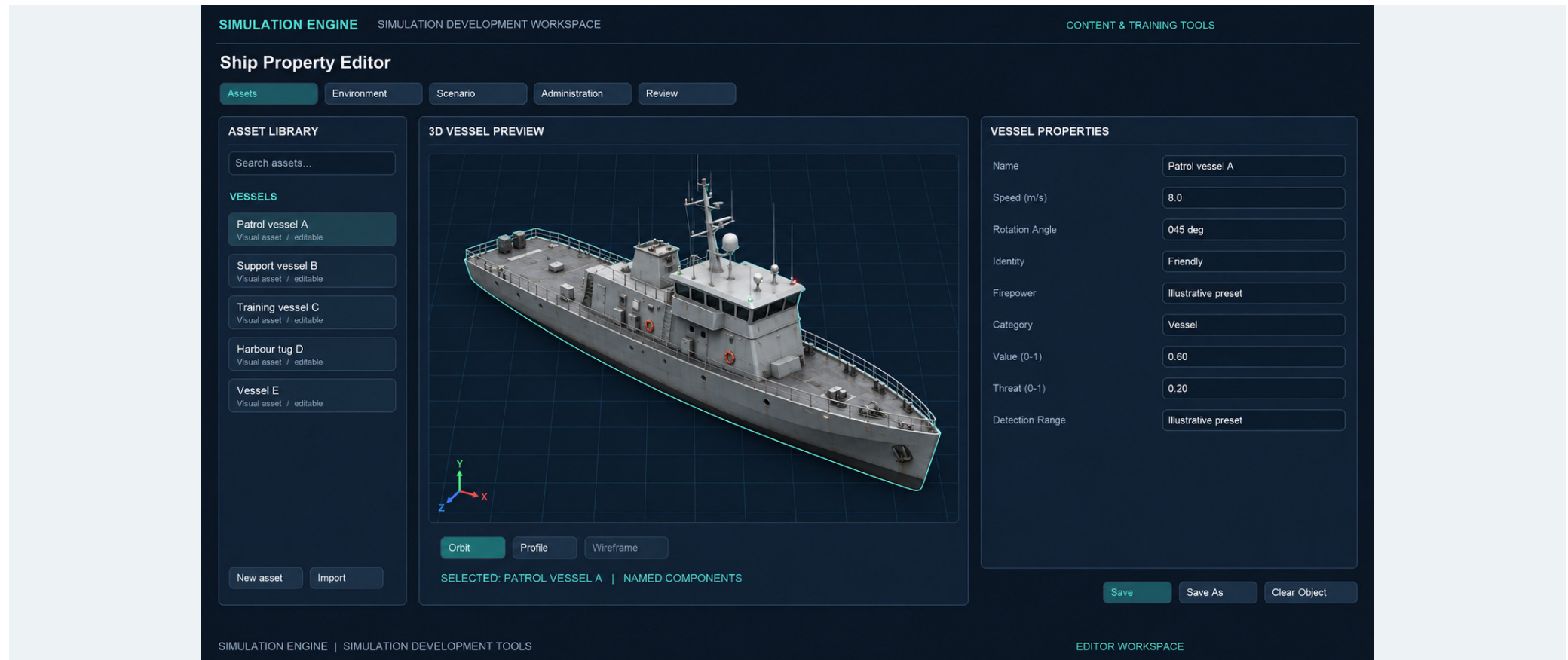
We processed a public reconstructed scan using Blender and Python, inspecting the mesh, deriving a vertex-point preview, simplifying geometry and exporting GLB. Face count falls from 935,255 to 139,968, approximately 85%, while the object remains recognisable.

Preparation for interactive content

Consistent camera views compare source reconstruction, vertex-point inspection and the optimised derivative. Our workflow prepares captured geometry for editing and interactive integration through mesh inspection, geometry reduction and asset export.

Scan inspection, mesh optimisation and GLB export. Source: Redwood Object Scans, Choi, Zhou, Miller and Koltun (2016), model 00021.

Ship Property Editor



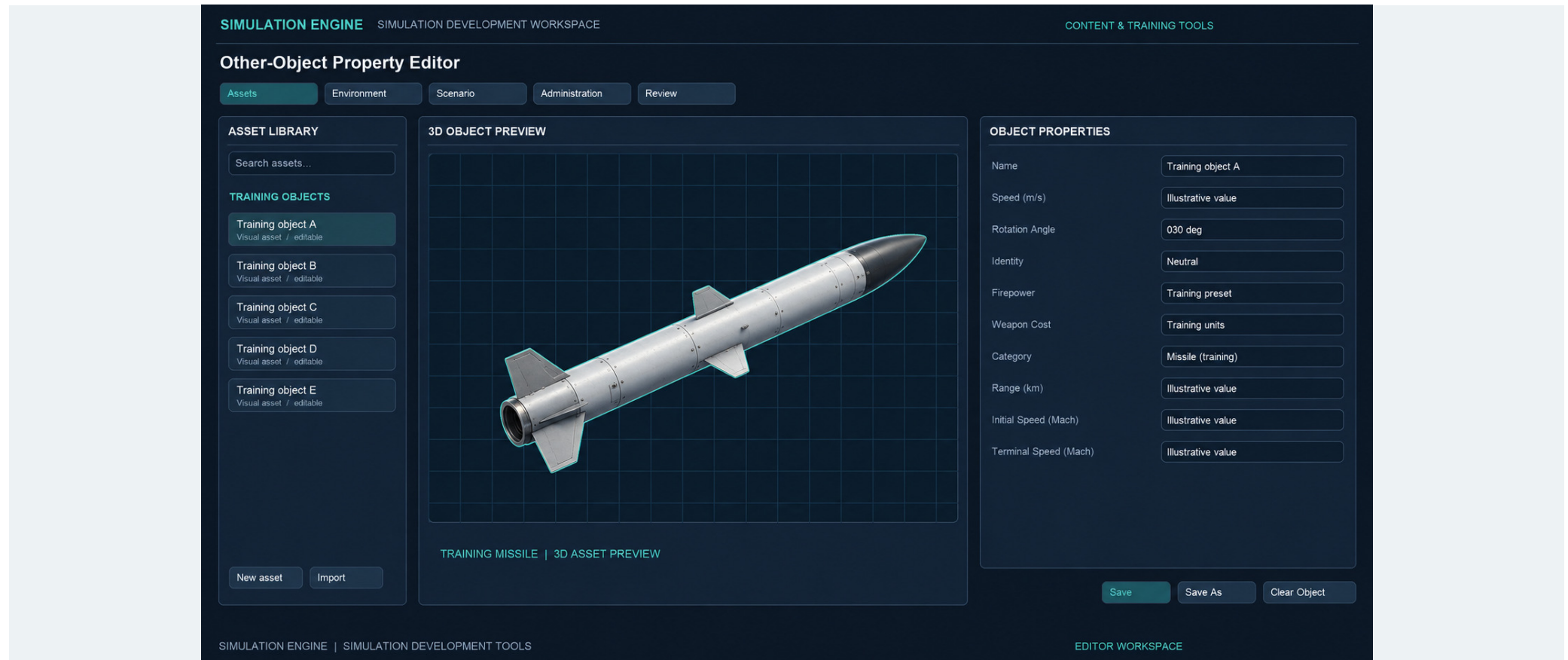
Vessel configuration

Our vessel editor workspace brings together asset selection, a 3D preview and configurable name, speed, rotation, identity and category. Vessel-specific fields include firepower, value, threat and detection range.

Content authoring workflow

Search, Save, Save As and Clear Object controls organise the authoring workflow. A selected-object outline and viewing controls provide a clear design for inspecting vessel content and configuring the training library.

Other-Object Property Editor



Category-specific configuration

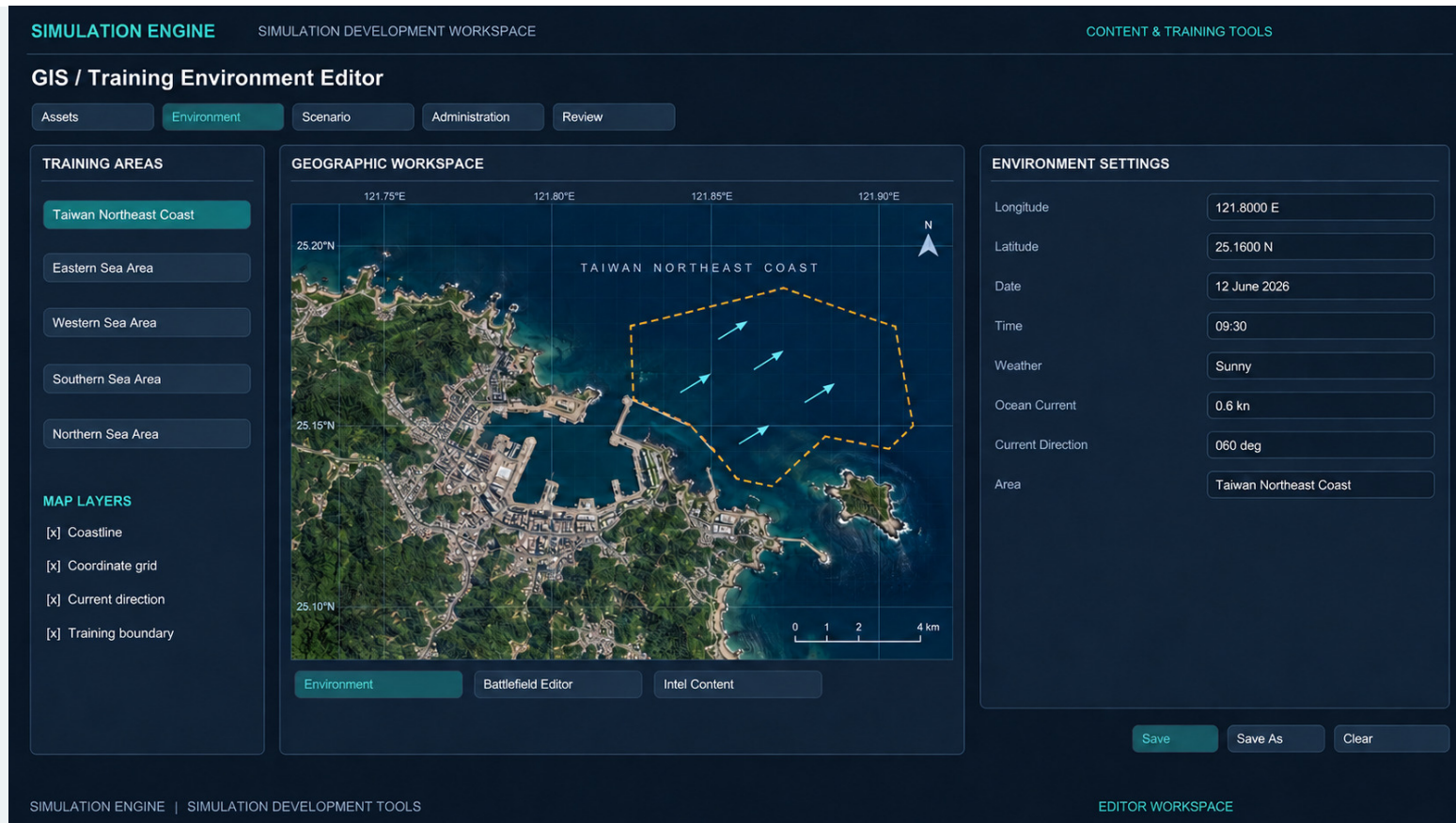
Our object editor design separates the asset library, geometry preview and parameter panel. The training-object example includes identity, rotation, speed, firepower, weapon cost, range and initial/terminal speed fields.

Reusable authoring structure

Search, selection and save controls are arranged within one editing workflow. Category-specific parameter panels organise the properties relevant to each object type, keeping reusable assets and training configuration together.

Other-Object Property Editor: asset selection and category-specific parameter configuration.

GIS and Training Environment Editor



Geographic exercise configuration

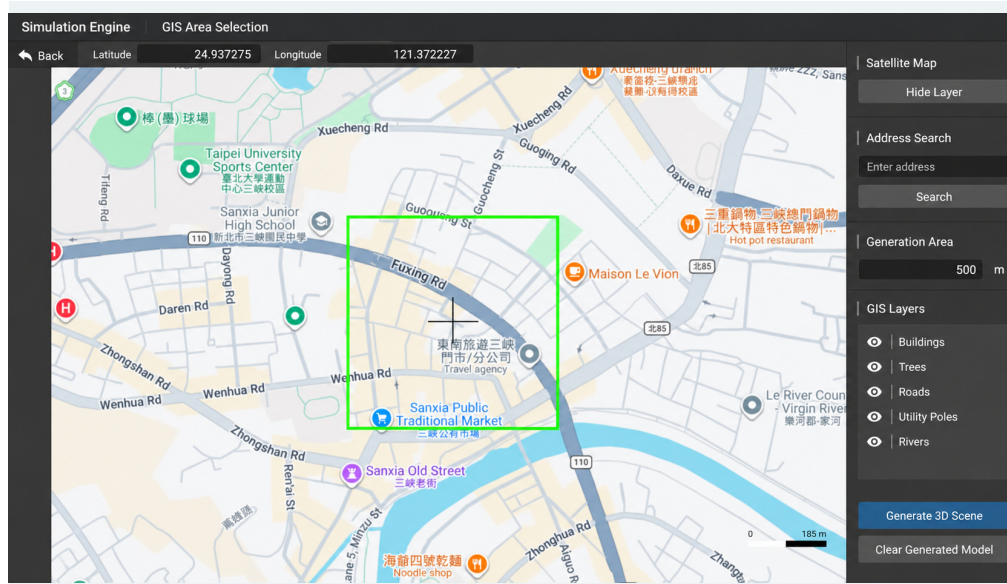
Our environment editor design combines regional area selection and a map workspace with longitude, latitude, date, time, weather and ocean-current settings. Layer controls and current arrows clarify the configured training environment.

Connected scenario tools

Environment, Battlefield Editor and Intel Content tabs connect geographic configuration with object placement and scenario information. Save, Save As and Clear organise the authoring workflow, bringing area settings and exercise content into one workspace.

GIS and Training Environment Editor: area selection, geographic layers, weather, time and ocean-current settings.

GIS data and 3D scene development

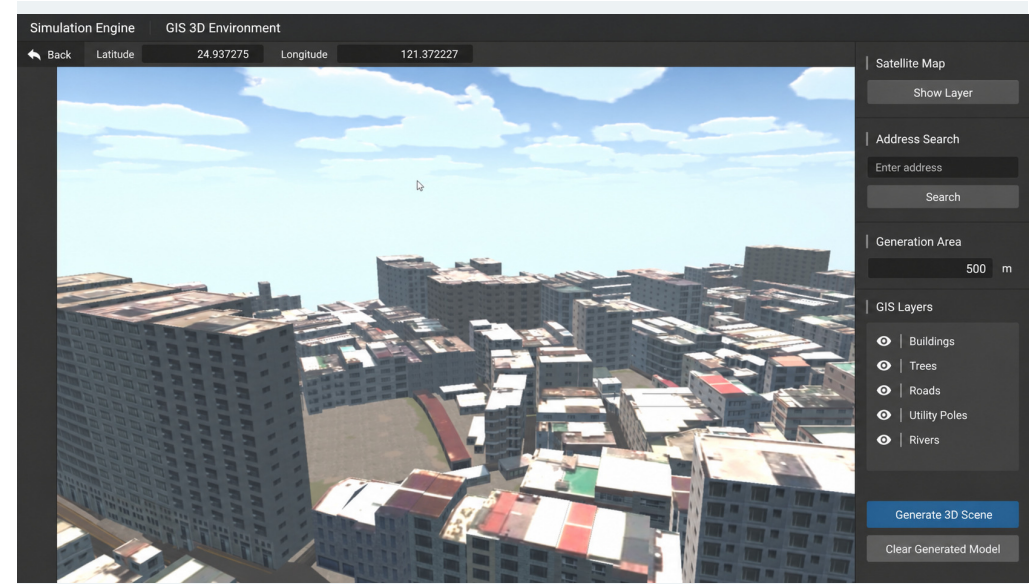


01 Select the geographic area

Integrate geographic data / Convert coordinates / Generate geometry / Align terrain / Edit the scene

Data integration and scene generation

Our GIS product combines public terrain and mapping data in Unity with a Cesium geographic foundation. Our development generates building geometry from 2D footprints and height attributes, converts TWD97 through WGS84 and ECEF to Unity coordinates, and applies height correction to align scene content with terrain.

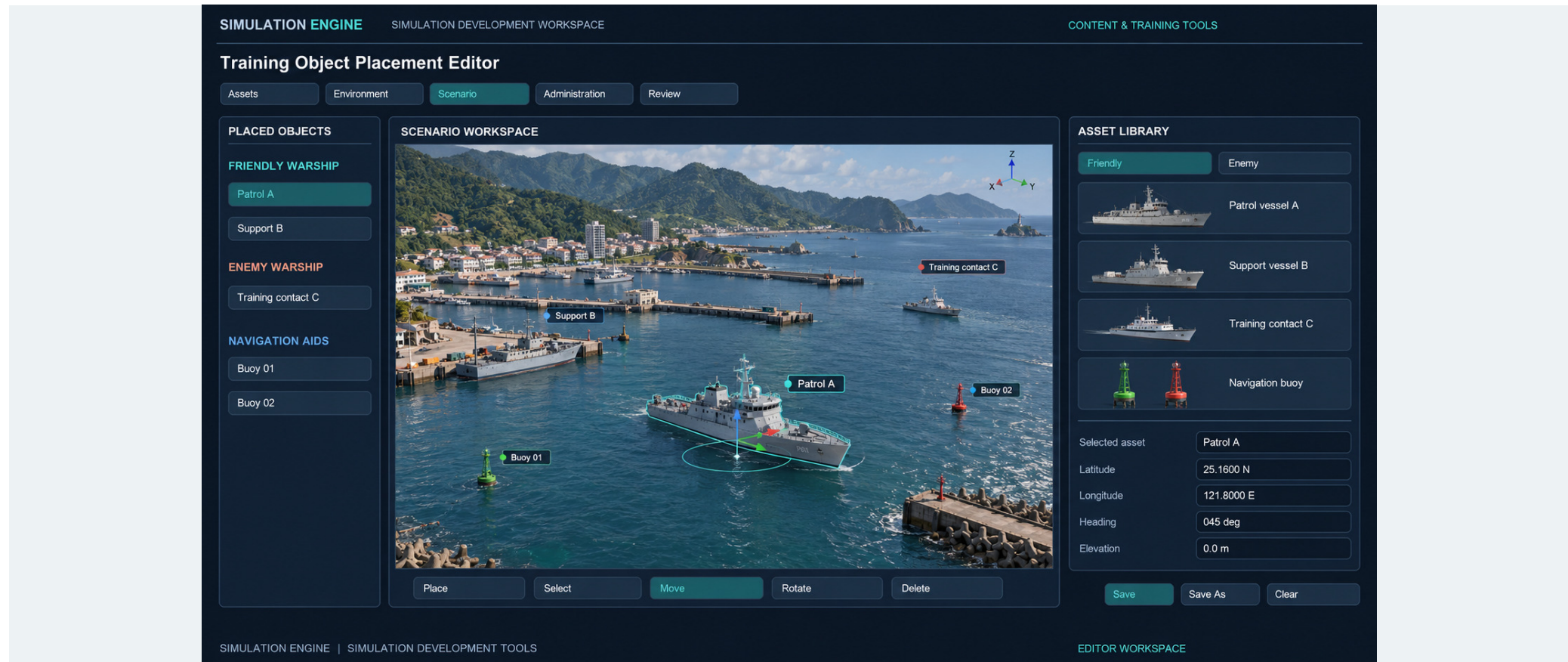


02 Generate the 3D environment

Editing and training application

Our product supports placement, translation and rotation, time and weather settings, and scenario/task authoring. Terrain, geographic layers and placed objects form reusable training-area definitions, connecting scenery development with structured content records and scenario configuration.

Training Object Placement Editor



Scenario composition

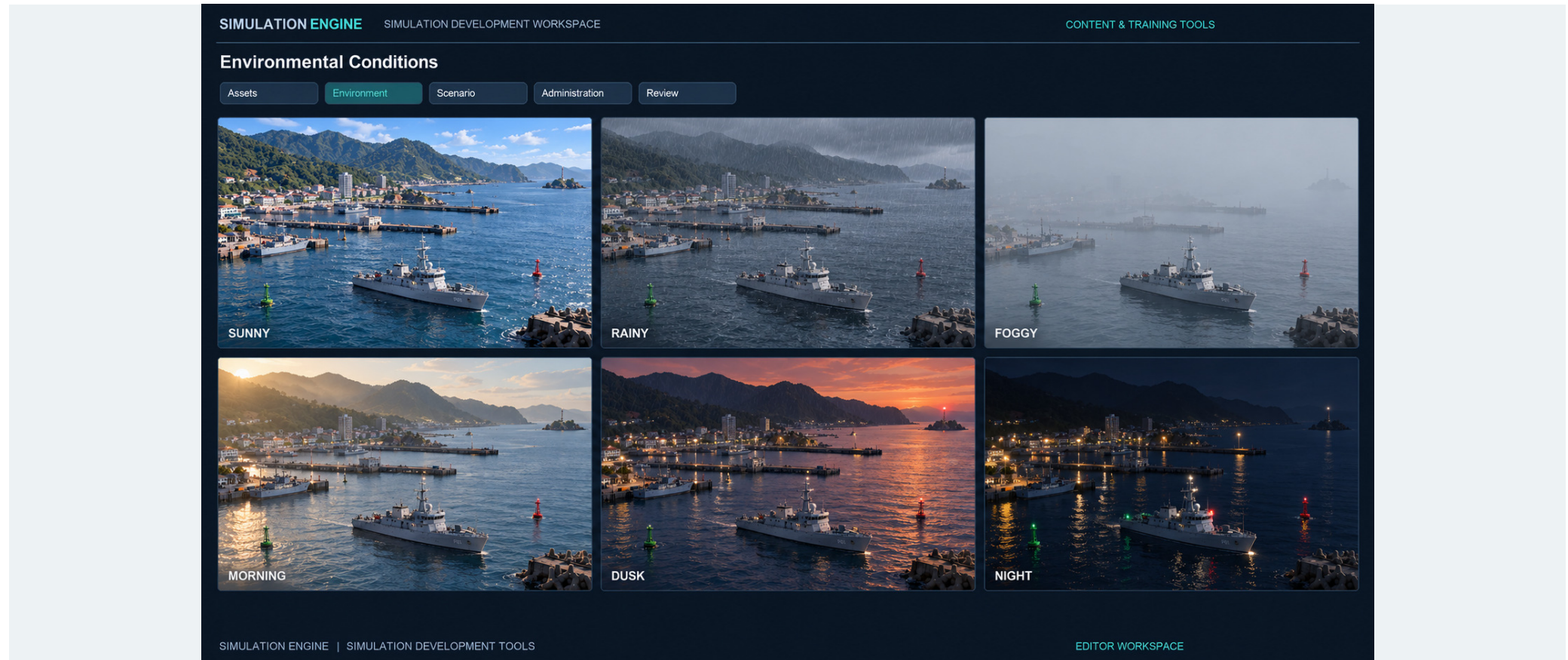
The placement workspace combines friendly/enemy object groups, an asset library and a selected vessel preview. Placement, selection, movement, rotation, deletion and saving controls organise scene assembly within a consistent interface design.

RMN content independence

Asset definition and scenario placement are presented as separate editing tasks. Our RMN delivery plan provides the agreed source content, developer tools and documentation to support independent scenario modification.

Training Object Placement Editor: scenario composition, object positioning and content library management.

Environmental conditions



Weather and lighting

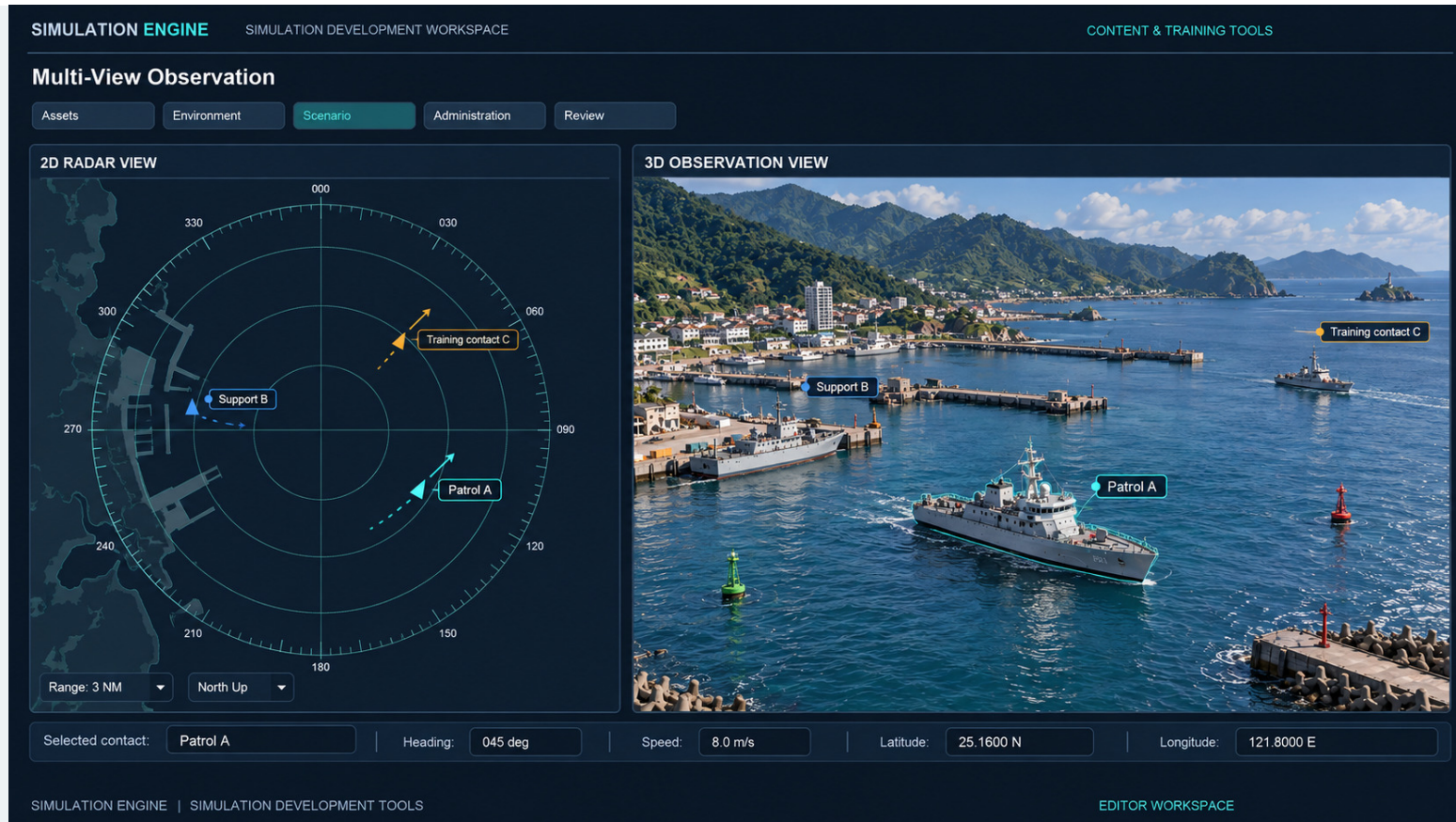
Sunny, rainy and foggy conditions sit alongside morning, dusk and night settings. These treatments show how a consistent training area can be presented across weather, visibility and lighting conditions.

Training application

Visibility and lighting variations support situational-awareness training. Our environment design connects presentation settings with the exercise context, enabling instructors to choose conditions suited to the training objective.

Environmental presentation: sunny, rainy, foggy, morning, dusk and night settings.

Multiple training views



Complementary displays

Our interface design presents a radar-style plan view beside a 3D observation window. Matching contact identities connect both displays, helping operators interpret relative position and the visual exercise scene together.

Coordinated display design

The display approach brings contact information, geographic context and observation views together. Configurable layouts support operator and instructor workspaces, with headings, speeds and positions arranged for clear exercise monitoring.

Coordinated radar-style and 3D observation views for exercise monitoring.

Training logs and review

SIMULATION ENGINE
SIMULATION DEVELOPMENT WORKSPACE

CONTENT & TRAINING TOOLS

Training Log & Review

Assets
Environment
Scenario
Administration
Review

RECORDING LIBRARY

12 JUNE 2026

Exercise 001 - Navigation

Exercise 002 - Visibility

Exercise 003 - Coordination

SELECTED RECORDING

Exercise 001

Duration 00:18:30

RECORDING PLAYBACK

00:07:20
00:07:50
00:08:10

00:08:10 / 00:18:30
Previous
Play / Pause
Next
1x speed

TRAINING EVENT / COMMAND LOG

TIME	ROLE	EVENT
00:07:20	Instructor	Weather preset updated: Sunny
00:07:50	Trainee	Course change acknowledged: 045 deg
00:08:10	Commander	Scenario checkpoint reviewed.

SIMULATION ENGINE | SIMULATION DEVELOPMENT TOOLS

EDITOR WORKSPACE

Exercise review workflow

The review workspace brings together date-organised recordings, selected playback, a timeline, playback controls and a command/event log. Exercise entries and timeline markers provide a structured way for instructors to revisit the training sequence.

Debrief and assessment

Playback and timestamped events support discussion of decisions and actions. The review design organises exercise history, commands and key moments for instructor-led debriefing and assessment workflows.

Training review: recording selection, timeline playback and timestamped command/event logs.

Scenario information and Intel Content

SIMULATION ENGINE
SIMULATION DEVELOPMENT WORKSPACE

CONTENT & TRAINING TOOLS

Scenario Information

Assets
Environment
Scenario
Administration
Review

FRIENDLY PLATFORM



Vessel name	Patrol A
Code name	DEMO-ALPHA
Identification	Friendly / Confirmed
Latitude	25.1600 N
Longitude	121.8000 E
Heading	045 deg
Speed	8.0 m/s

CONFIGURATION DETAILS

Visual model	Patrol vessel A
Behaviour preset	Harbour transit
Launcher configuration	Training preset
Firepower summary	Training parameters

TRAINING CONTACT



Vessel name	Training contact C
Code name	DEMO-CHARLIE
Identification	Training contact / Confirmed
Latitude	25.1750 N
Longitude	121.8150 E
Heading	225 deg
Speed	6.0 m/s

CONFIGURATION DETAILS

Visual model	Training vessel C
Behaviour preset	Coastal transit
Launcher configuration	Training preset
Firepower summary	Training parameters

SCENARIO LOCATION MAP



Environment
Battlefield Editor
Intel Content

SIMULATION ENGINE | SIMULATION DEVELOPMENT TOOLS
EDITOR WORKSPACE

Platform summaries

The scenario information design separates friendly platforms from training contacts. Vessel thumbnails, name, code name, coordinates, heading, speed and configuration fields organise platform information for the instructor.

Connected authoring workflow

Environment, placement and scenario information share a consistent editing structure. Platform summaries connect names, coordinates and configuration with the geographic workspace, giving instructors a clear overview of exercise content.

Scenario information: platform summaries, coordinates, configuration and geographic context.

User and role management

The screenshot displays the 'User and Role Management' interface within the 'SIMULATION ENGINE' workspace. The interface is divided into several sections:

- Navigation:** Assets, Environment, Scenario, Administration (active), Review.
- USER DIRECTORY:** A table listing users with columns for NAME, ACCOUNT, ROLE, and STATUS. A search bar and a dropdown for 'All roles' are present.
- ACCOUNT SETTINGS:** A form for configuring an account, including fields for ID, Name, Account, Password, and Status.
- ROLE SELECTION:** Radio buttons for selecting a role: Coach, Trainee, or Commander.
- SELECTED ROLE ACCESS:** Checkboxes for permissions: Scenario preparation, Exercise control, and Session review.
- ROLE ACCESS SUMMARY:** A table summarizing the permissions for each role.

USER DIRECTORY Table:

NAME	ACCOUNT	ROLE	STATUS
Instructor 01	demo.coach	Coach	Active
Trainee 01	demo.trainee	Trainee	Active
Commander 01	demo.commander	Commander	Active

ACCOUNT SETTINGS Form:

ID: DEMO-COACH-01
Name: Instructor 01
Account: demo.coach
Password: *****
Status: Active

ROLE SELECTION: Coach (selected), Trainee, Commander

SELECTED ROLE ACCESS: Scenario preparation, Exercise control, Session review

ROLE ACCESS SUMMARY Table:

ROLE	DESCRIPTION
Coach	Prepare scenarios, control exercises, review sessions
Trainee	Join assigned exercises, view own session records
Commander	Coordinate assigned exercises, review team sessions

Account administration

The account administration design brings together a user directory, ID, account and masked-password fields, coach/trainee/commander roles, and Add, Edit and Delete controls.

Role-based training organisation

Role selection and access summaries organise scenario preparation, exercise participation and session review. The workspace presents account settings and role permissions together for clear training administration.

User administration: account configuration, role selection and access management.

Simulation content database and management

SIMULATION ENGINE v2

CONTENT DEVELOPMENT WORKSPACE

Content database

Development tools

Simulation Content Database and Management

Structured assets · Editable properties · Revision history · Export and restore

Ships
1

Training areas
1

Other objects
1

Scenarios
1

CONTENT LIBRARY

Search records

DEV_SHIP_001
ship · revision 2

HARBOUR_001
scenery · revision 1

BUOY_001
object · revision 1

SCENARIO_001
scenario · revision 1

Export catalogue

Restore catalogue

ASSET DEFINITION

Identifier

DEV_SHIP_001

Category

Ship

Display name

Patrol vessel - grey material configuration

Asset / configuration reference

ship_development/DEV_SHIP_001.glb

Structured properties

```
{  
  "asset_format": "GLB",  
  "source_project": "DEV_SHIP_001.blend",  
  "material_set": "naval_grey"  
}
```

Check record

Save revision

REVISION HISTORY

Revision	Record name
v1	Patrol vessel - exterior asset
v2	Patrol vessel - grey material configuration

CONTENT LIFECYCLE

Keep asset references and configuration together. Retain changes as revisions and exchange catalogue records through a structured JSON file.

Persistent catalogue

Required-field checks

Structured property checks

Validated catalogue restore

Create → Edit → Check → Save revision → Export → Restore

VR UNIVERSAL SDN. BHD. | SHIPS · AREAS · OBJECTS · SCENARIOS

Structured simulation content

Our catalogue workspace organises ship, training-area, object and scenario definitions through unique identifiers, asset references and structured properties. Category selection, search and record editing keep content information in one workspace.

Revision and recovery workflow

The working catalogue retains revision history, checks record fields and properties, and exchanges records through JSON export and restore. Our Python/SQLite processing example adds database integrity checks, release manifests and backup/recovery routines.

Development tools and RMN content workflow

SIMULATION ENGINE v2

CONTENT DEVELOPMENT WORKSPACE

Content database

Development tools

RMN Development Tools and Content Workflow

Simulation Engine v2 development scope · Tools, editable formats and operator outcomes

WORKSPACE / TECHNOLOGY	DEVELOPMENT FUNCTION	EDITABLE CONTENT	RMN WORKFLOW
Ship authoring Blender + Python	Geometry, materials and asset optimisation	BLEND / GLB	Edit vessel appearance and prepare runtime assets
GIS scene generation Unity + Cesium	Geographic foundation and generated scene content	Layer references / Scene configuration	Select an area, generate scenery and edit placement
Vessel & object editors Simulation Engine v2	Asset selection and category-specific properties	Asset references / Parameter records	Configure vessel and training-object definitions
Environment & scenario editors Simulation Engine v2	Weather, area settings, objects and scenario composition	JSON scenario definitions	Create, modify and load training exercises
Content catalogue JavaScript / JSON	Record editing, revision history and catalogue exchange	JSON records and revisions	Create records, check fields, export and restore
Database processing Python / SQLite	Structured storage, integrity checks and recovery	SQLite database / Release manifest	Inspect revisions, retain backups and recover content

Author

 →

Configure

 →

Maintain

 →

Package

 →

Recover

TECHNOLOGY FOUNDATIONS: BLENDER · PYTHON · UNITY · CESIUM · JAVASCRIPT · SQLITE

Tools connected to tasks

The tool inventory connects vessel authoring, GIS scene generation, property editing, scenario configuration and content maintenance. Editable geometry, asset references, parameter records and scenario definitions organise the development workflow around RMN content tasks.

Developer enablement

Technical documentation and practical worksheets guide model export, area configuration, record maintenance and content recovery. Our technology scope spans Unity, Unreal Engine, 3D authoring, GIS, programming, databases and integration, detailed in the following technology pages.

Simulation Engine v2 development scope: tool functions, editable formats and RMN workflows. SOC alignment: 2.7.1.6.3.2 and 3.4.4.

Real-time engines and simulation capabilities

VR UNIVERSAL / TECHNOLOGY AND DEVELOPMENT

Real-time simulation and immersive systems

Unity

C# / Unity workflows

Simulation scenes, interactive tools, training logic and VR applications.

Unreal Engine

C++ / Blueprints

Real-time visualisation, environment systems and interactive simulation.

Simulation physics

Buoyancy / wind / currents

Ship response, propulsion, rudder behaviour and sea-state configuration.

VR and physical interaction

OpenXR / device integration

Tracked viewpoints, input devices, controls and immersive training.

Instructor and scenarios

Scenario logic / telemetry

Exercise configuration, object placement, monitoring and replay workflows.

Rendering and optimisation

LOD / materials / profiling

Scalable scene detail, lighting, textures and runtime performance tuning.

AUTHOR > CONFIGURE > INTEGRATE > TEST > DELIVER > MAINTAIN

Real-time development

Our technology stack combines Unity and C# with Unreal Engine, C++ and Blueprints for interactive simulation, visualisation and immersive training. Engine workflows cover scene construction, training logic, editor tools, materials, lighting and performance optimisation.

Maritime and immersive systems

Development capabilities span ocean conditions, ship motion, buoyancy, propulsion, rudder control, wind and currents. VR interaction, physical controls, instructor functions, scenario configuration and telemetry connect the simulation experience to training tasks.

3D software, capture and GIS production

VR UNIVERSAL / TECHNOLOGY AND DEVELOPMENT

3D production, geographic areas and editable assets

3D modelling and CAD

Blender / 3ds Max / Maya

Mesh authoring, ship geometry, hard-surface modelling and CAD conversion.

Surface and material work

Substance 3D / Photoshop

PBR textures, surface detail, material variants and texture optimisation.

Capture and reconstruction

Photogrammetry / 3D scans

Reference capture, alignment, mesh cleanup, retopology and UV mapping.

GIS and terrain

Cesium / QGIS / GDAL

Geographic layers, terrain preparation, coordinates and area content.

Asset interchange

BLEND / FBX / GLB / OBJ

Editable source files, runtime exports, asset references and versioned delivery.

Training-area construction

Terrain / ports / landmarks

Coastlines, harbour structures, buoys, scenery and reusable scene content.

AUTHOR > CONFIGURE > INTEGRATE > TEST > DELIVER > MAINTAIN

Editable 3D production

Our 3D workflow covers Blender, 3ds Max and Maya, supported by Substance 3D and Photoshop for textures and materials. Capabilities include ship and object modelling, CAD conversion, UV mapping, retopology, material variants and low-to-high detail asset preparation.

Capture and geographic content

Photogrammetry and 3D scan processing connect reference capture to cleaned, editable models. Unity/Cesium, QGIS and GDAL support geographic layers, terrain, coordinate conversion and training-area preparation, with FBX, GLB and OBJ supporting asset exchange.

Languages, databases and software engineering

VR UNIVERSAL / TECHNOLOGY AND DEVELOPMENT

Programming, databases and system integration

Engine programming

C# / C++ / Blueprints

Simulation behaviours, editor extensions, device interfaces and runtime logic.

Automation and processing

Python / SQL

Asset processing, geographic conversion, database tasks and batch workflows.

Web and content tools

JavaScript / HTML / CSS

Editable catalogues, dashboards, property forms and browser interfaces.

Data and storage

SQLite / JSON / CSV

Structured records, scenarios, revision history, manifests and data exchange.

Integration and networking

APIs / TCP / UDP

Simulator components, instructor stations, device input and state exchange.

Engineering and handover

Git / testing / documentation

Version control, checks, packaging, backup, deployment and developer training.

AUTHOR > CONFIGURE > INTEGRATE > TEST > DELIVER > MAINTAIN

Programming and data tools

C#, C++ and Blueprints support engine behaviours and editor development. Python automates asset and geographic processing; SQL supports database workflows. JavaScript, HTML and CSS support editable catalogues and interfaces, while SQLite, JSON and CSV organise records and scenario data.

Integration and maintainability

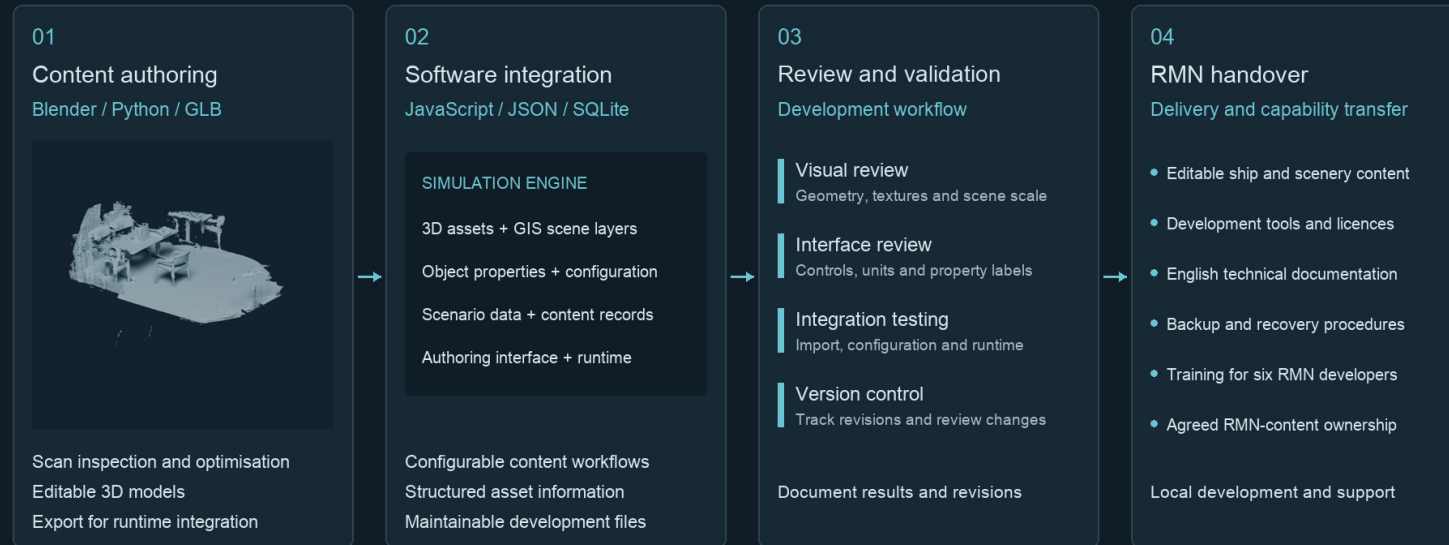
System development covers APIs, TCP/UDP communication, device input and simulator component integration. Git version control, repeatable checks, runtime profiling, release packaging, backups and technical documentation support maintainable in-house development and RMN developer handover.

Local development and RMN independence

VR UNIVERSAL / DEVELOPMENT CAPABILITY

From editable content to maintainable simulations

In-house authoring and integration methods supporting the proposed Malaysian RMN delivery



Malaysian delivery approach: design, programming, integration, testing and technical support.

Development tools, technical documentation and training support independent content maintenance.

Development tools and Malaysian delivery

Our technology scope connects Unity and Unreal Engine, editable 3D assets, GIS production, C#/C++ programming, Python automation and structured databases. Malaysian delivery brings design, programming, testing, integration and technical support together across ships, training areas and content tools.

Editable content and capability transfer

Our RMN delivery approach includes editable ship and scenery content, development tools, licence inventories, English documentation, backup/recovery and training for six RMN developers. Content ownership, technical handover and capability transfer support independent scenario maintenance and future content development.

Simulation development. Local capability.

Editable content, connected training systems and development tools for RMN.

3D & GIS CONTENT

Vessel modelling, scan processing, geographic scene generation and asset optimisation.

SIMULATION & TRAINING

Immersive interaction, instructor workspaces, scenario authoring and exercise review.

LOCAL DEVELOPMENT

Software integration, technical documentation, developer training and lifecycle support.